

How do we get that

$$I_\beta(t) = \int_{\mathbb{R}^d} \frac{1}{(1 + |\xi|^2)^{1-\alpha}} \left(1 - e^{-t(|\xi|^2 + \beta)}\right) \hat{f}(d\xi)$$

When I use mathematica, I get that

$$I_\beta(t) = \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{(1 + |\xi|^2)^{1-\alpha}} \left[\Gamma(1 - \alpha) - \Gamma(1 - \alpha, t(\beta + |\xi|^2)) \right].$$

I used the following in mathematica to find the dr integral:

`Integrate[r^(-a) * Exp[-r *( z^2 + b)], {r,0,t}, Assumptions-> a>0 && z>0 && b>0 && t>0]`

Le's answer: What I have done is let $|\xi| = 0$ and ∞ to obtain the lower and upper bounds, namely,

$$\left(1 - e^{-\beta t}\right) \Upsilon_\alpha(\beta) \leq I_\beta(t) \leq \Upsilon_\alpha(\beta)$$

Please let me know if this is correct?

[Nick \(Nov 26 2022\)](#): I am still unclear on how you are showing that

$$I_\beta(t) = \int_{\mathbb{R}^d} \frac{1}{(1 + |\xi|^2)^{1-\alpha}} \left(1 - e^{-t(|\xi|^2 + \beta)}\right) \hat{f}(d\xi).$$

All I am able to show is the following:

$$\begin{aligned} I_\beta(t) &= \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) e^{-r(|\xi|^2 + \beta)} \\ &= \int_{\mathbb{R}^d} \hat{f}(d\xi) \int_0^t dr r^{-\alpha} e^{-r(|\xi|^2 + \beta)} \\ &= \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{(|\xi|^2 + \beta)^{1-\alpha}} \int_0^{t(|\xi|^2 + \beta)} dr r^{-\alpha} e^{-r} \end{aligned}$$

where the last line is from, the substitution $r = r(|\xi|^2 + \beta)$. Thus by denoting $\Gamma(s, x) = \int_x^\infty t^{1-s} e^{-t} dt$, this is equivalent to

$$I_\beta(t) \leq \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{(|\xi|^2 + \beta)^{1-\alpha}} (\Gamma(1 - \alpha) - \Gamma(1 - \alpha, t(|\xi|^2 + \beta)))$$

Using this, I am having trouble getting a bound equivalent to

$$\left(1 - e^{-\beta t}\right) \Upsilon_\alpha(\beta) \leq I_\beta(t) \leq \Upsilon_\alpha(\beta).$$

The upper bound is easy,

$$(\Gamma(1 - \alpha) - \Gamma(1 - \alpha, t(|\xi|^2 + \beta))) \leq \Gamma(1 - \alpha),$$

and thus

$$I_\beta(t) \leq \Gamma(1 - \alpha) \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{(|\xi|^2 + \beta)^{1-\alpha}} = \Gamma(1 - \alpha) \Upsilon_\alpha(\beta).$$

However, I am unable to show the lower bound as $(\Gamma(1 - \alpha) - \Gamma(1 - \alpha, t(|\xi|^2 + \beta))) \rightarrow 0$ as $|\xi| \rightarrow 0$.

Let's Answer: for the lower bound, notice that $1 - e^{-t(|\xi|^2 + \beta)} \geq 1 - e^{-t\beta}$. Hence,

$$\begin{aligned}
I_\beta(t) &= \int_{\mathbb{R}^d} \frac{1}{(1 + |\xi|^2)^{1-\alpha}} \left(1 - e^{-t(|\xi|^2 + \beta)}\right) \hat{f}(\mathrm{d}\xi) \\
&\geq \left(1 - e^{-t\beta}\right) \int_{\mathbb{R}^d} \frac{1}{(1 + |\xi|^2)^{1-\alpha}} \hat{f}(\mathrm{d}\xi) \\
&= \left(1 - e^{-t\beta}\right) \Gamma(1 - \alpha) \Upsilon_\alpha(\beta).
\end{aligned}$$